Alireza Bahremand

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SUMMARY

XR engineer with 7 years of hands-on Unity/OpenXR development — currently building XR teleoperation and data-capture pipelines that teach humanoid robots, from headset tracking through low-latency streaming into robot control. Shipped VR products serving **1,000+ weekly users**; co-inventor on **3 U.S. patents** in mixed-reality systems.

TECHNICAL SKILLS

- **XR & Spatial Computing:** Unity (XR Interaction Toolkit, XR Hands, Timeline, URP, VFX Graph, Editor tooling), OpenXR, Meta SDKs (Quest Passthrough Camera API, Depth API), VITURE XR SDK (Carina VIO, native dylib bridges), ARKit/ARFoundation, RealityKit/visionOS, Vive OpenXR eye tracking, Unreal Engine, NVIDIA Omniverse
- **Tracking, Streaming & Vision:** real-time head/hand tracking (OVRSkeleton, MediaPipe), SLAM/VIO (Kimera-VIO, MASt3R), volumetric video streaming/playback (point clouds, Draco compression, NTP sync, 4D Gaussian splats), WebRTC, depth estimation (Depth Anything 3, Depth Pro), OpenCV, on-device inference (ONNX, Sentis)
- **Robotics:** XR teleoperation for simulated humanoids (Unitree H1/G1, Fourier GR-1) and physical Reachy Mini hardware, analytic FK/IK, sim-to-sim VLA fine-tuning (NVIDIA Isaac-GR00T), MuJoCo, WebSocket/LCM/UDP control protocols
- **Languages & Systems:** C, C++, C#, Python, Swift; native plugin/dylib bridges, FastAPI/REST, GPU infrastructure (NVIDIA DGX Spark, Lambda Cloud), CI/CD, multiplayer networking (Photon Fusion, LAN/WAN)

RELEVANT EXPERIENCE

Chaotic Curiosity LLC — Research Engineer

May 2025 – Present

- Build XR teleoperation clients for capturing humanoid-robot training demonstrations in simulation: a **Meta Quest 3** app (Passthrough Camera API, metric head/hand tracking via OVRSkeleton) and a **VITURE Luma Ultra** smart-glasses path (native Carina VIO dylib bridge + MediaPipe hand tracking), each streaming pose over low-latency UDP/WebSocket into an analytic IK chain driving simulated **Unitree H1/G1 and Fourier GR-1** humanoids — the glasses pipeline runs live, with real head/hand tracking driving the control chain.
- Built a robot-agnostic demonstration-recording pipeline — raw episode capture with calibration fingerprinting, offline re-retargeting to new robot layouts (verified bit-exact on same-robot replay), and an automated episode-QA gate — hardening the XR-capture-to-training-data path.
- Built pipelines to fine-tune vision-language-action foundation models (**NVIDIA Isaac-GR00T N1.7**) for sim-to-sim transfer — retraining simulated humanoid robots against new simulated environments, closing the loop from XR-captured demonstrations to learned robot behavior.
- Integrated custom hardware with off-the-shelf XR devices: stereo passthrough + 6DoF pose on VITURE glasses fused with MASt3R / Depth Anything 3 reconstruction for real-time dense geometry, plus 4D Gaussian-splat playback on-device.
- Led a VR behavioral-research system integrating **eye-tracking, motion-capture, telemetry, and BCI hardware**, building the full data collection and playback pipeline.

Dreamscape Learn (EdPlus at ASU) — VR Programmer

December 2023 – Present

- Built and shipped three educational VR courses supporting **1,000+ weekly active student sessions** — implementing features, fixing bugs, and keeping a production XR fleet reliable at scale.
- Led VR classroom deployment/testing infrastructure (computer vision + LAN/WAN multiplayer) and automated build/deploy pipelines with agent-based test automation — an estimated **$100K+** in avoided infrastructure and deployment costs, while mentoring student developers.

Arizona State University — Research Assistant / PhD Student

August 2018 – June 2023

- Lead researcher on an adaptive **volumetric video streaming** system — real-time capture, compression, and low-latency playback — presented at **ACM MobiSys**, adopted by 2+ research teams; mentored 5+ students.
- Built the **Smell Engine**, an NSF-funded olfactory VR platform integrating custom scent-delivery hardware with off-the-shelf VR headsets in real time (**IEEE VR**; basis for U.S. patent 12,653,920).

NASA Marshall Space Flight Center — XR Software Engineering Intern

Summer 2019

- Led a team of 4 building a VR training-authoring tool and a CAD-import pipeline; used it to create two ISS maintenance simulations, presented to the MSFC board as a cost-reduction strategy.

EDUCATION

Arizona State University — M.S. Computer Engineering, 2021 (GPA 3.8) • B.S. Software Engineering, 2018 (GPA 3.9)

SELECTED ACHIEVEMENTS

- Co-inventor on **3 U.S. patents** in mixed-reality systems: fluid simulation, illumination estimation, and olfactory delivery (US 12,653,920 B2)
- **Meta Mixed Reality Hackathon Winner** (2024) • ASU President's Innovation Award (2024) • NVIDIA AI Certified • **5+ peer-reviewed publications** (ACM MobiSys, IEEE VR, ACM HotEdgeVideo)